

PHY 614-2011
Problem Set #10

Problem 1 (12.5.12)

$$x_1 = r \sin \theta \sin \phi$$

$$x_2 = r \sin \theta \cos \phi$$

$$x_3 = r \cos \theta$$

Under $\theta \rightarrow \pi - \theta$

$$\cos \theta \rightarrow -\cos \theta$$

$$\sin \theta \rightarrow \sin \theta$$

Under $\phi \rightarrow \phi + \pi$

$$\cos \phi \rightarrow -\cos \phi$$

$$\sin \phi \rightarrow -\sin \phi$$

Thus $\theta \rightarrow \pi - \theta$ and $\phi \rightarrow \phi + \pi$

implies $x_1 \rightarrow -x_1, x_2 \rightarrow -x_2, x_3 \rightarrow -x_3$

The spherical harmonics are

$$Y_\ell^m(\theta, \phi) = A_{\ell m} e^{im\phi} P_\ell^m(\cos \theta)$$

where $A_{\ell m} \rightarrow$ Normalization const

$$Y_\ell^m(\theta, \phi + \pi) = A_{\ell m} e^{im(\phi + \pi)} P_\ell^m(-\cos \theta)$$

First consider $m=l$

Then from eqn (12.5.32)

$$\begin{aligned}\pi Y_l^l(\theta, \phi) &= Y_l^l(\pi - \theta, \phi + \pi) \\ &= (-1)^l \left[\frac{(2l+1)!}{4\pi} \right]^{1/2} \frac{1}{2^l l!} (\sin \theta)^l e^{il\phi} e^{il\pi} \\ &= (-1)^l Y_l^l(\theta, \phi)\end{aligned}$$

Now consider the operator

$$L_- = -\hbar e^{-i\phi} \left(\frac{\partial}{\partial \theta} - i \cot \theta \frac{\partial}{\partial \phi} \right)$$

Under $\theta \rightarrow \pi - \theta$ $\phi \rightarrow \phi + \pi$

$$\frac{\partial}{\partial \theta} \rightarrow -\frac{\partial}{\partial \theta}, \quad \cot \theta \rightarrow -\cot \theta, \quad \frac{\partial}{\partial \phi} \rightarrow \frac{\partial}{\partial \phi}$$

$$e^{-i\phi} \rightarrow -e^{-i\phi}$$

Thus L_- does not change

$$\Rightarrow [\pi, L_-] = 0$$

$$\begin{aligned}\text{Thus } \pi Y_l^m &\sim \pi (L_-)^{l-m} Y_l^l \\ &\sim L_-^{l-m} \pi Y_l^l \\ &\sim (-1)^l Y_l^m\end{aligned}$$

Problem 2 (12.6.1)

$$\psi = A e^{-r/a_0}$$

(1) Since ψ is independent of θ, ϕ , $l = m = 0$

$$\begin{aligned} (2) \quad & \frac{1}{r^2} \frac{d}{dr} r^2 \frac{d}{dr} \psi \\ &= \frac{1}{r^2} \frac{d}{dr} r^2 \left[-\frac{A}{a_0} e^{-r/a_0} \right] \\ &= -\frac{A}{a_0} \frac{1}{r^2} \frac{d}{dr} (r^2 e^{-r/a_0}) \\ &= -\frac{A}{a_0} \frac{1}{r^2} \left[2r - \frac{1}{a_0} r^2 e^{-r/a_0} \right] \\ &= \left(-\frac{2A}{a_0 r} + \frac{A}{a_0} \right) e^{-r/a_0} \end{aligned}$$

Thus Schrödinger eqn becomes

$$-\frac{\hbar^2}{2\mu} \left(\frac{1}{a_0} - \frac{2}{r a_0} \right) + V(r) = E$$

Assuming $V(r) \rightarrow 0$ as $r \rightarrow \infty$, the leading term as $r \rightarrow \infty$ in LHS is the first one. This shows $E = -\frac{\hbar^2}{2\mu a_0^2}$

$$(3) \Rightarrow V(r) = -\frac{\hbar^2}{\mu a_0 r}$$

Problem 3 (12.6.4)

(1) By definition

$$\delta^{(3)}(\vec{r} - \vec{r}') = 0 \quad \vec{r} \neq \vec{r}'$$

This is clearly satisfied by the
RHS

Furthermore by definition

$$\int d^3r \delta^{(3)}(\vec{r} - \vec{r}') = 1$$

Using RHS

$$\int d^3r \frac{1}{r^2 \sin \theta} \delta(r-r') \delta(\theta-\theta') \delta(\phi-\phi')$$

$$= \int dr d\theta d\phi \frac{r^2 \sin \theta}{r^2 \sin \theta} \delta(r-r') \delta(\theta-\theta') \delta(\phi-\phi')$$

$$= \left(\int d\theta \delta(\theta-\theta') \right) \left(\int d\phi \delta(\phi-\phi') \right) \left(\int dr \delta(r-r') \right)$$

$$= 1$$

is satisfied

(2) In polar coordinates

$$\nabla^2 f = + \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \right]$$

When $f = 1/r$

$$\nabla^2 \left(\frac{1}{r} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \right) = 0 \quad \text{unless } r=0$$

Now consider

$$\int d^3 r \nabla^2 \left(\frac{1}{r} \right) = \oint d\vec{s} \vec{\nabla} \cdot \left(\frac{1}{r} \right)$$



The integral on RHS is over a small sphere of radius r_0 around $r=0$

$$\text{RHS} = 4\pi r_0^2 \left(-\frac{1}{r_0^2} \right) = -4\pi$$

$$\Rightarrow \int d^3 r \nabla^2 \left(\frac{1}{r} \right) = -4\pi$$

Since $\nabla^2 \left(\frac{1}{r} \right) = 0$ for $\vec{r} \neq 0$

$$\nabla^2 \left(\frac{1}{r} \right) = -4\pi \delta^{(3)}(\vec{r})$$

Problem 4 (12.68)

For $l=0$, the function $U_0(r)$ satisfies

$$\frac{d^2 U_0}{dr^2} + \frac{2\mu}{\hbar^2} [E - V(r)] U_0 = 0$$

where
$$V(r) = \begin{cases} 0 & r < r_0 \\ \infty & r \geq r_0 \end{cases}$$

Then for $r < r_0$ the solution is $(k^2 = 2\mu E/\hbar^2)$

$$U_0(r) = A \cos kr + B \sin kr.$$

Since the boundary condition at $r=0$ requires $U(0)=0$, we must have $A=0$

Since $U_0(r_0)=0$

$$\sin kr_0 = 0$$

ie

$$k = \frac{n\pi}{r_0} \quad n = 0, \pm 1, \pm 2$$

$$\Rightarrow E = \frac{\hbar^2 \pi^2 n^2}{2\mu r_0^2}$$

Problem 5 (12.6.9)

The $l=0$ eqn for $U_0(r)$ is for $E < 0$

$$\frac{d^2 U_0}{dr^2} + \frac{2\mu}{\hbar^2} [E - V(r)] U_0 = 0$$

$$V(r) = \begin{cases} -V_0 & \text{for } r < r_0 \\ 0 & \text{for } r > r_0 \end{cases}$$

General soln for $r < r_0$ which satisfies condition $U_0(0) = 0$ is

$$U_0^{(I)}(r) = A \sin k' r \quad r < r_0$$

where $k'^2 = \frac{2\mu(E + V_0)}{\hbar^2}$

and for $r > r_0$ Since $U_0(\infty)$ has to vanish

$$U_0^{(II)}(r) = B e^{-kr} \quad r > r_0$$

where $k^2 = \frac{2\mu|E|}{\hbar^2}$

Matching at $r = r_0$ means

$$U_0^{(I)}(r_0) = U_0^{(II)}(r_0)$$

$$U_0'^{(I)}(r_0) = U_0'^{(II)}(r_0)$$

Thus

$$A \sin k' r_0 = B e^{-k r_0}$$

$$k' A \cos k' r_0 = -k B e^{-k r_0}$$

These have a solution if

$$\frac{1}{k'} \tan k' r_0 = -\frac{1}{k}$$

$$\text{or} \quad \tan k' r_0 = -\frac{k'}{k}$$

The radial part of wave fn

$$\psi = R_l(r) Y_l^m(\theta, \phi)$$

$$R_l(r) = \frac{U_l(r)}{r}$$

Satisfies for $l=0$

$$\frac{d^2 U_l}{dr^2} + \frac{2\mu}{\hbar^2} [E + A e^{-r/a}] U_l = 0$$

where

$$\mu = \frac{m_p m_n}{m_p + m_n} \simeq \frac{1}{2} m$$

With

$$\xi = e^{-r/2a}$$

This becomes to

$$\frac{d^2 U_l}{d\xi^2} + \frac{1}{\xi} \frac{dU_l}{d\xi} + \left(c^2 - \frac{k^2}{\xi^2} \right) U_l = 0$$

where

$$c^2 = \frac{8\mu}{\hbar^2} A a^2$$

$$k^2 = - \frac{8\mu E a^2}{\hbar^2}$$

The general solution is

$$U_l(\xi) = A J_k(c\xi) + B J_{-k}(c\xi)$$

Boundary conditions are

$$U_l(r=0) \approx r=0 \Rightarrow U_l(\xi=1) = 0$$

$$U_l(r=\infty) \rightarrow 0 \Rightarrow U_l(\xi=0) = 0.$$

Since $J_v(x) \sim x^v$, the condition
 $U_0(0) = 0 \Rightarrow B = 0$.

The other boundary condition
then gives

$$A J_k(c) = 0$$

Thus $J_k(c) = 0$

This gives a relation between
 k and c

ie between E and (A, a)